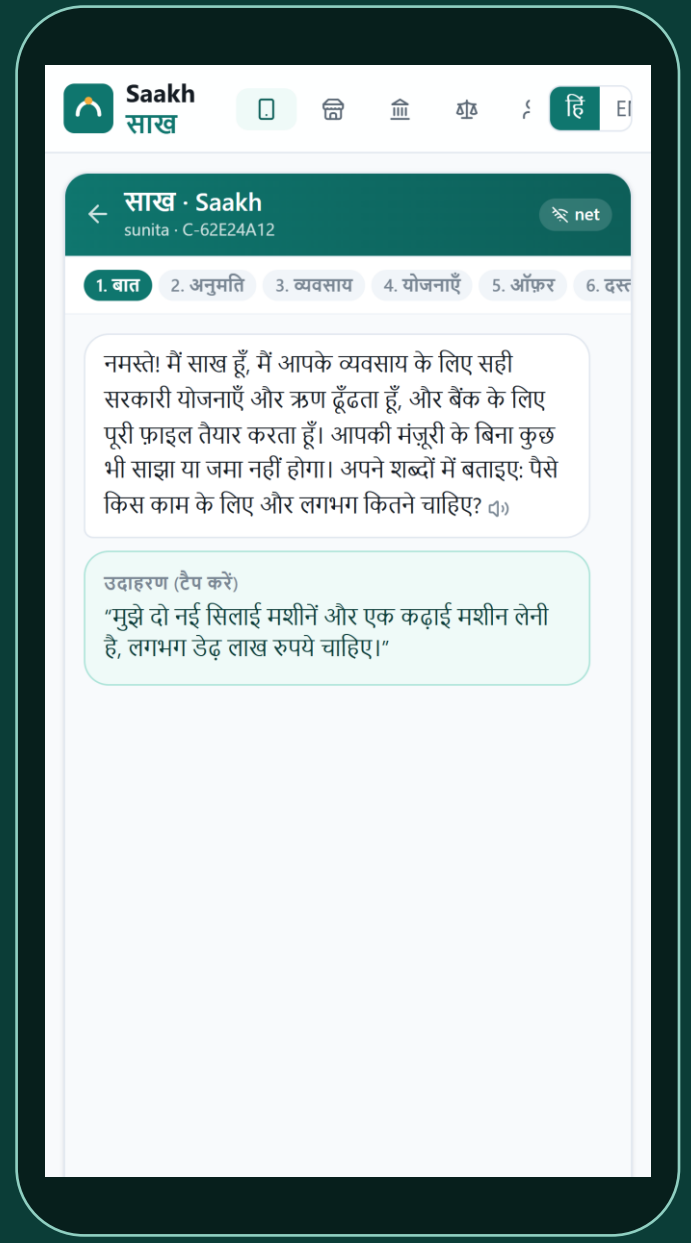
TECH4BHARAT 2026 · PROBLEM 3.2 FORMAL CREDIT AND SCHEMES · AGENTIC AI

# Saakh साख

*From UPI trail to bank-ready credit*

An AI agent that helps India's invisible micro enterprises reach formal credit and the government schemes built for them, with their consent, a cited source for every claim, and their approval before anything is sent.




**Team Soliloquy**
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# Viable businesses, invisible to banks

## Meet Sunita, a tailor in Barabanki

- Stitches for about 60 paying customers a month, mostly paid by UPI
- No ITR, so the bank cannot see her
- Qualifies for PM Vishwakarma and MUDRA; nobody has told her
- Repays a moneylender ₹1,200 every Monday on a ₹50,000 loan, about 57% a year

*The barrier is not creditworthiness. It is paperwork, scheme complexity and months of branch visits.*

**₹30 lakh cr**

addressable MSME credit gap, about 24% of demand

*SIDBI with Crisil, May 2025*

**7.9 crore+**

enterprises on the Udyam platforms, about 99% of them micro

*PIB, May 2026*

**24 to 36%**

interest paid to moneylenders when formal credit fails

*Tech4Bharat problem brief*

**15 to 20**

papers per loan today, plus a 10 to 15% agent cut

*Problem brief; to verify in the field*

# One agent, about two minutes, a bank-ready file

1

## Talk

Hindi, Hinglish or English, by voice

2

## Consent

4 separate, purpose-bound artefacts

3

## Understand

Cash flow, seasons, moneylender loans

4

## Match

10 scheme pathways, every rule cited

5

## Ask

Only questions that change an outcome

6

## Offer

A loan sized to lean and busy months

7

## Build

17 papers become 9 items, DPR generated

8

## Approve

The owner approves the exact file

9

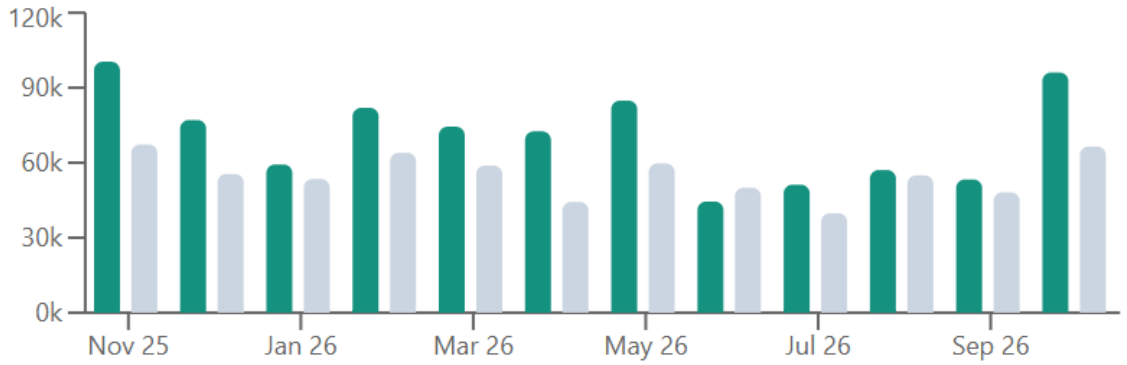
## Track

The branch gets a pre-appraised file

*Built on India's digital rails: UPI, Account Aggregator, DigiLocker, Udyam. In the prototype these are simulated adapters with production-shaped interfaces, labelled SIMULATED.*

# It reads the business, not the paperwork

From your account (AA) 12 months SIMULATED




**Average business income ₹71,043/month.** 93% received digitally from 60 paying customers a month.

**Seasonal business.** Peak: Nov, Oct, May. Lean: Jun, Jul, Sep (lean month ≈ 60% of a normal month).

**Likely moneylender repayments.** 52 weekly payments of ₹1,200 to RAMKUMAR YADAV (≈ ₹62,400/year). Refinancing this is part of the plan.

## Moneylender found

52 weekly payments of ₹1,200 to one person: flagged as an informal loan, about 57% a year, and refinanced in the offer.

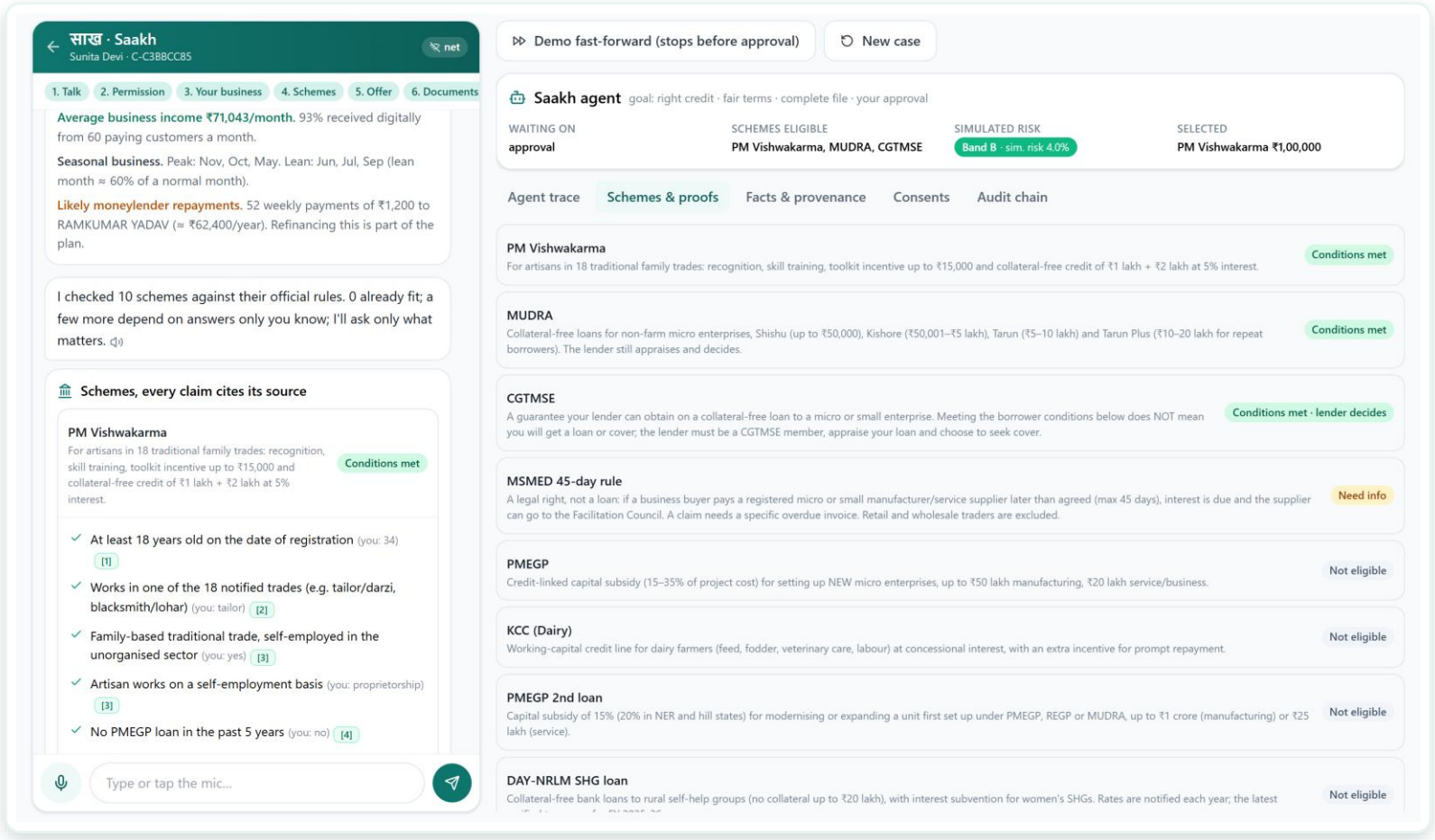
## Seasons, not averages

Lean months run at about 60% of normal, so the repayment plan shrinks when income does.

## Evidence, minimal data

Turnover, B2B buyers and bounces are derived with evidence. Account Aggregator data is processed in memory and never stored.

# Every scheme claim comes with its source




**Conditions met**

every criterion passes, each with its citation

**Provisional**

rests on a self-declaration an agency verifies

**Needs info**

the agent asks the one most decisive question

**Lender decides**

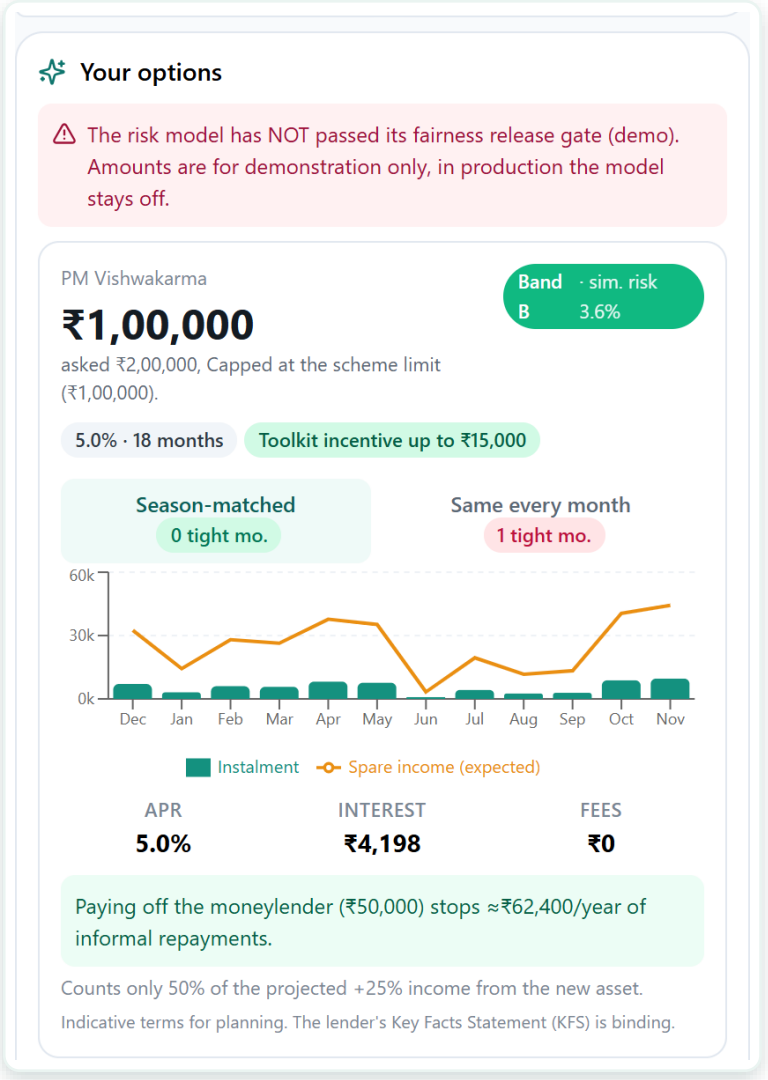
e.g. CGTMSE cover is the bank's choice

**Not available**

e.g. Stand-Up India, period ended March 2025

*10 scheme pathways · rules are data (YAML) · the loader refuses any rule without a citation · unknowns never become "eligible"*

# A loan sized to the seasons, not a fixed EMI

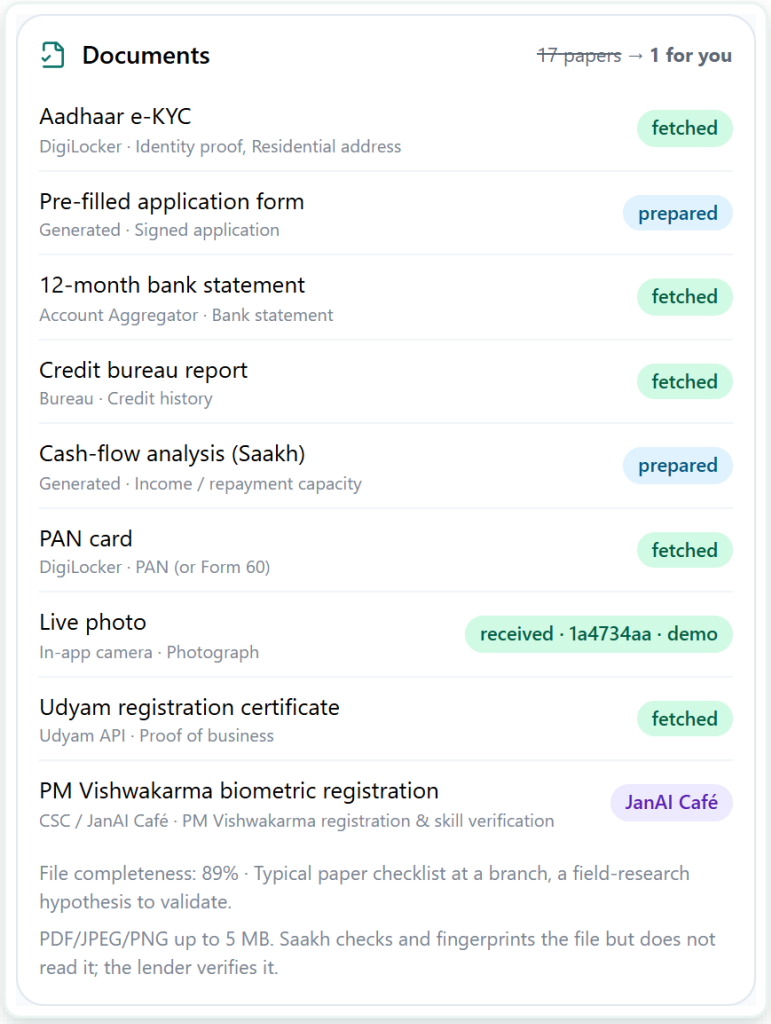


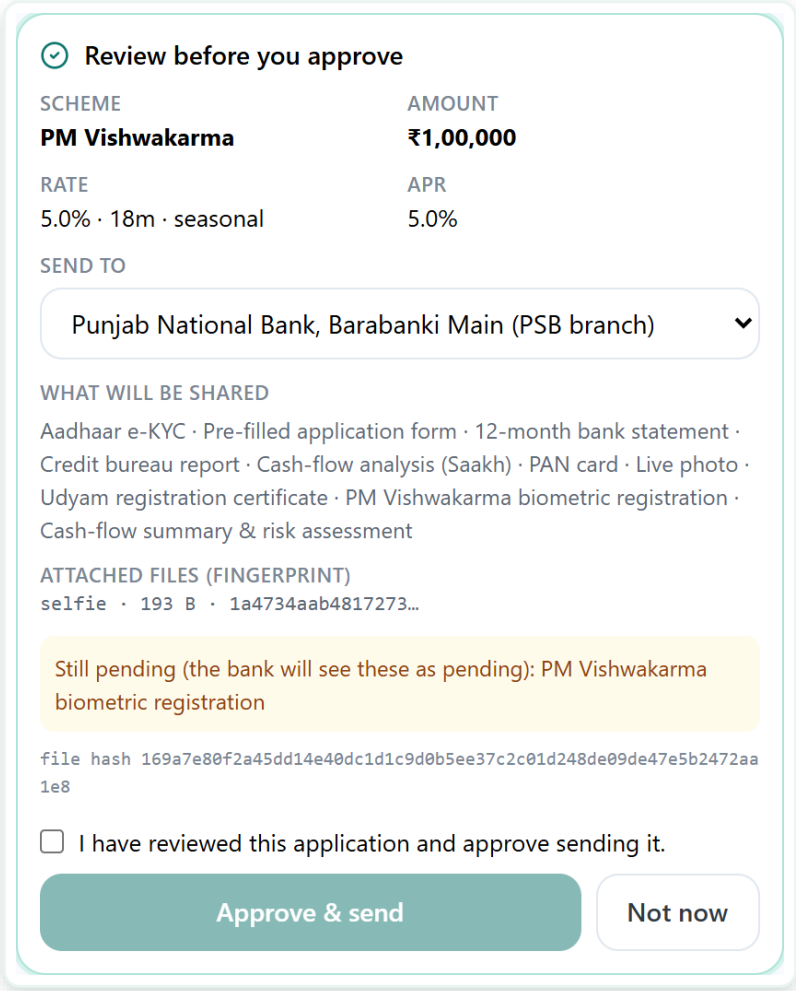

# 12 → 0

tight months for a dairy unit over a 36-month loan: same-every-month EMI vs a season-matched schedule with the same present value

- Sized from disposable cash flow and risk policy, with only half of the projected uplift from the new asset
- Key Facts Statement-style APR, interest and fees, shown before choosing
- Refinances the moneylender: for Sunita, it stops about ₹62,400 a year of informal repayments
- An honest "not now" with a path back, when a loan would strain the business

# From 17 papers to 9 items, approved as an exact file




## Minimum-effort set cover

Most items are fetched or generated: pre-filled form, cash-flow analysis, project report PDF. Physical steps go to a JanAI Café.

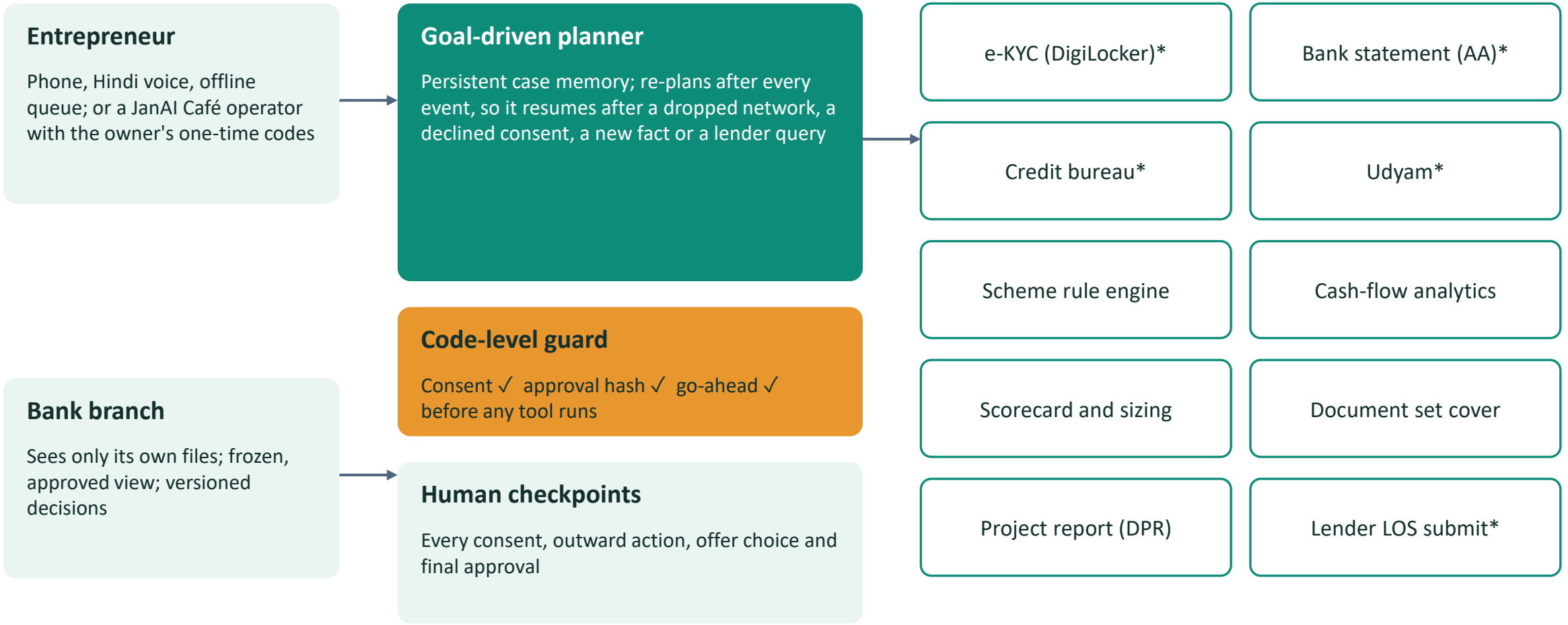
## Bound to a fingerprint

Approval covers the SHA-256 of the file and of every attachment. Any change voids it; a sent copy is superseded until re-approved.

## Revocable

Withdrawing consent withdraws the file and erases its contents from the bank's copy.

# A real agent: plans, acts, recovers, and asks the human




*Recovery: the bank-data call times out by design; the agent retries with backoff, then falls back to a statement upload or a café visit. * simulated adapters. Eligibility, risk and amounts come from auditable engines, never from generated text.*

# Trust is enforced in code, not promised

### Purpose-bound consent

ReBIT-style artefacts with expiry. Decline any one and the agent asks instead.

### Revocation that deletes

Withdrawing or expiry erases derived data, chat cards and the bank's copy, on a schedule.

### Tamper-evident audit

A hash-chained log that stores no personal or financial values.

### Exact-file approval

SHA-256 binds the file and attachments; café codes are one-time, per step, 10-minute.

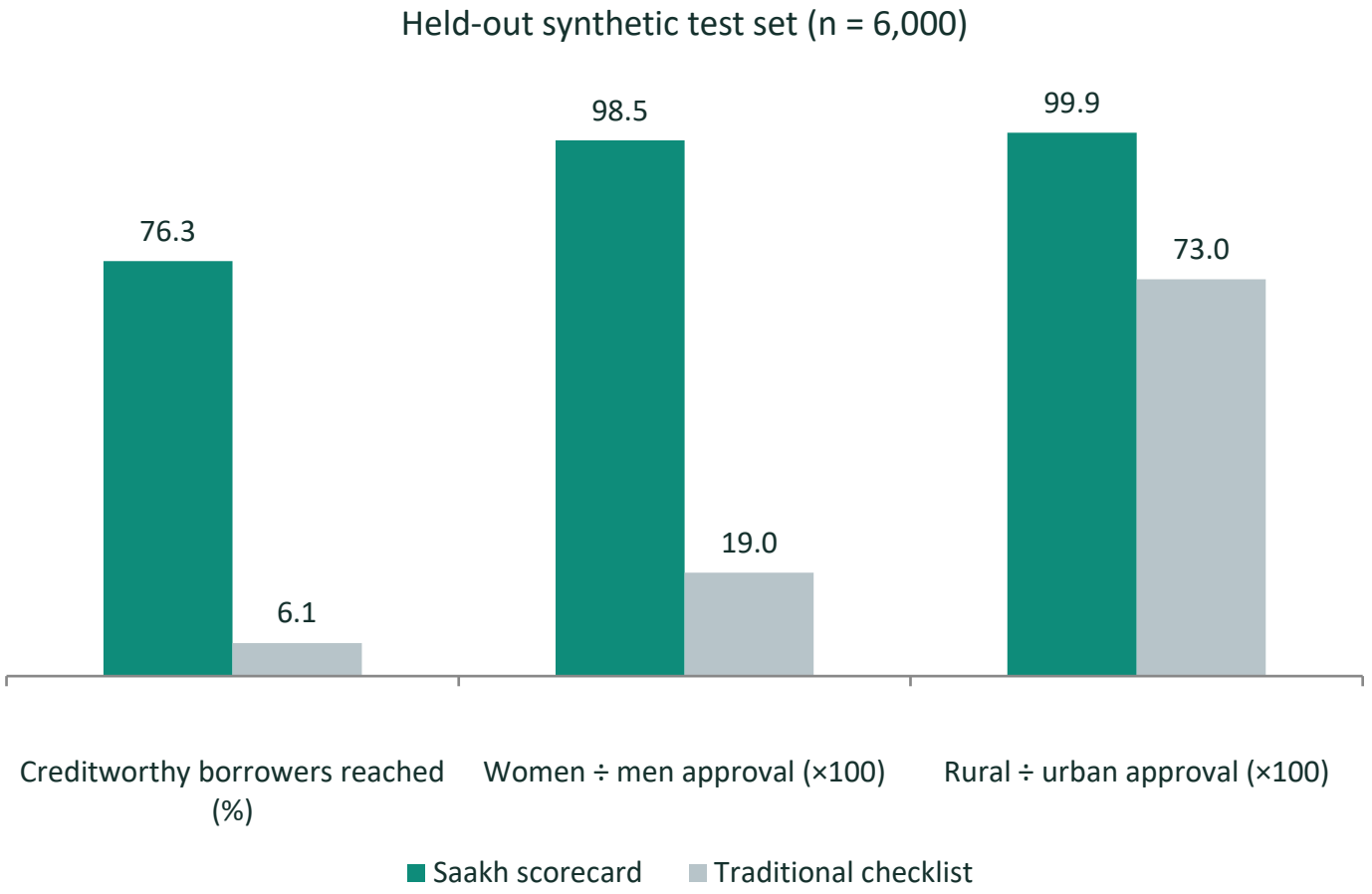
### Lender isolation

Each branch sees only its files; decisions are versioned, final and idempotent.

### Strict inputs

Every event is type-checked; actions are accepted only at their checkpoint.

# Fairer than the checklist, and honest when a gate fails



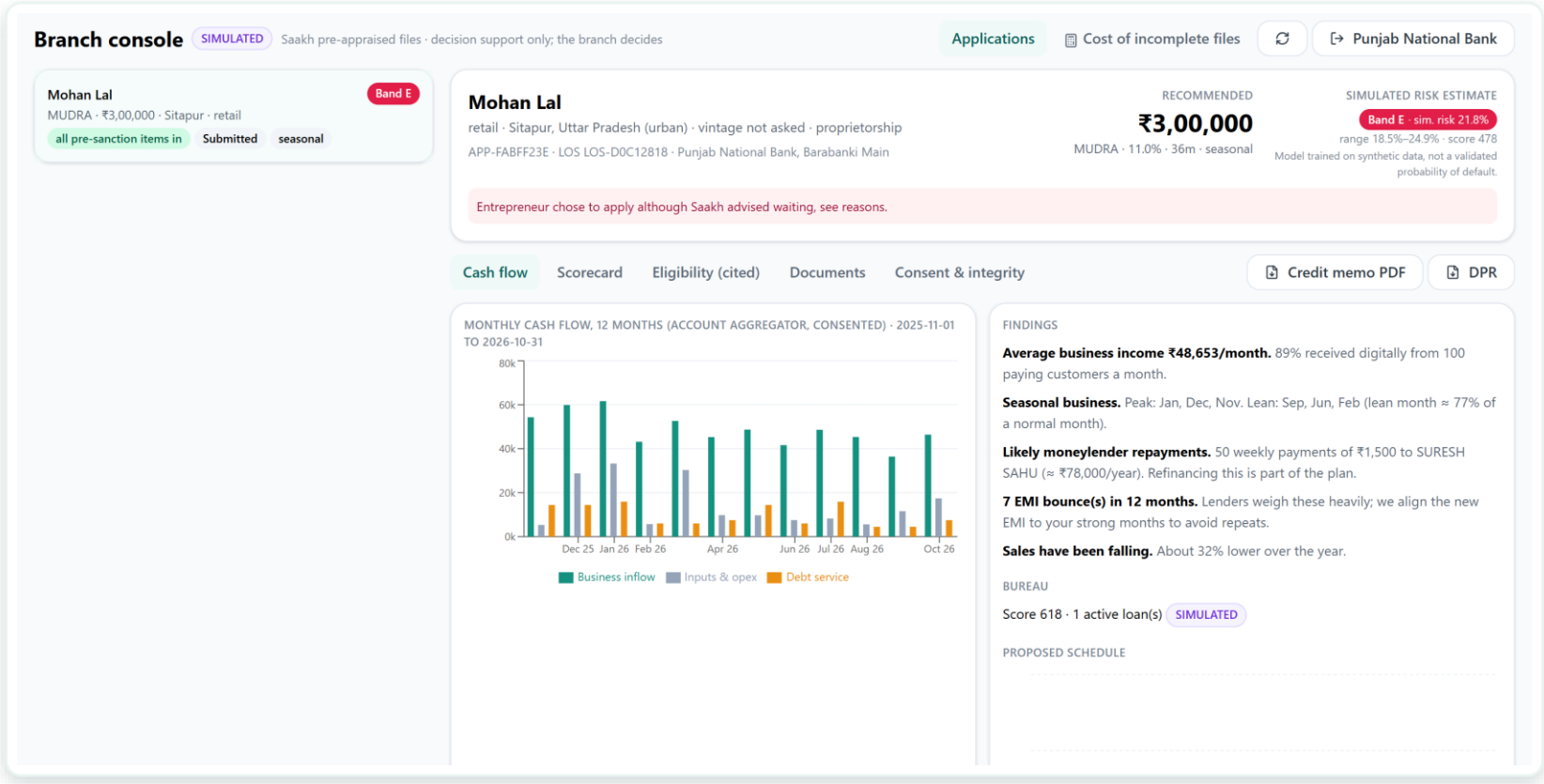

## BLOCKED, by design

1 of 7 fairness gates fails (ST vs General, −7.3 pp on the test split; −1.85 pp on a fresh draw). The rule fails closed: production refuses score-based offers until the gap is explained.

- Monotone, explainable points table; reason codes for every score
- Gender, caste, religion and location are never inputs
- AUC 0.736, calibration error 0.6%; a boosted-tree challenger gains nothing

*Labels come from a synthetic generator: these numbers show the pipeline works, not real-world performance. Lender data is needed before any real use.*

# For the bank: a pre-appraised file it can trust




## ≈ ₹79,000

rework cost avoided per branch per month (about 105 officer-hours)

- Credit memo and project report PDFs
- Reason codes, cited eligibility, integrity proofs
- Fingerprint-checked document downloads
- Declines need the bank's own reasons

*Rework estimate uses editable assumptions (60 files a month, 45% incomplete today vs 5%, 2.2 rework cycles of 1.6 hours, ₹750 per officer-hour) to be replaced with field measurements.*

# Free for the owner, paid for by those who save



## Entrepreneur

**Pays nothing**

Removes the 10 to 15% agent cut and months of visits.



## Banks and NBFCs

**₹150 to 400 per complete file + success fee**

Rework alone costs about ₹1,300 per file today.



## DICs and scheme agencies

**Programme and outreach budgets**

Cohort screening helps them meet PMEGP and PM Vishwakarma targets.



## SHG federations, BCs, JanAI Cafés

**Assisted-service fee per journey**

Screen 40 members in seconds; earn per completed file.

*Near-zero marginal cost per file, so growth is bounded by distribution, not compute.*

# Large market, national rails, local jobs

**₹30 lakh cr**

MSME credit gap

**7.9 cr+**

Udyam enterprises

**~99%**

of them micro

## Go to market

1. Pilot: one bank branch and one SHG federation
2. Regional rollout with that lender
3. National: AA, ULI, DigiLocker rails and the CSC / JanAI Café network

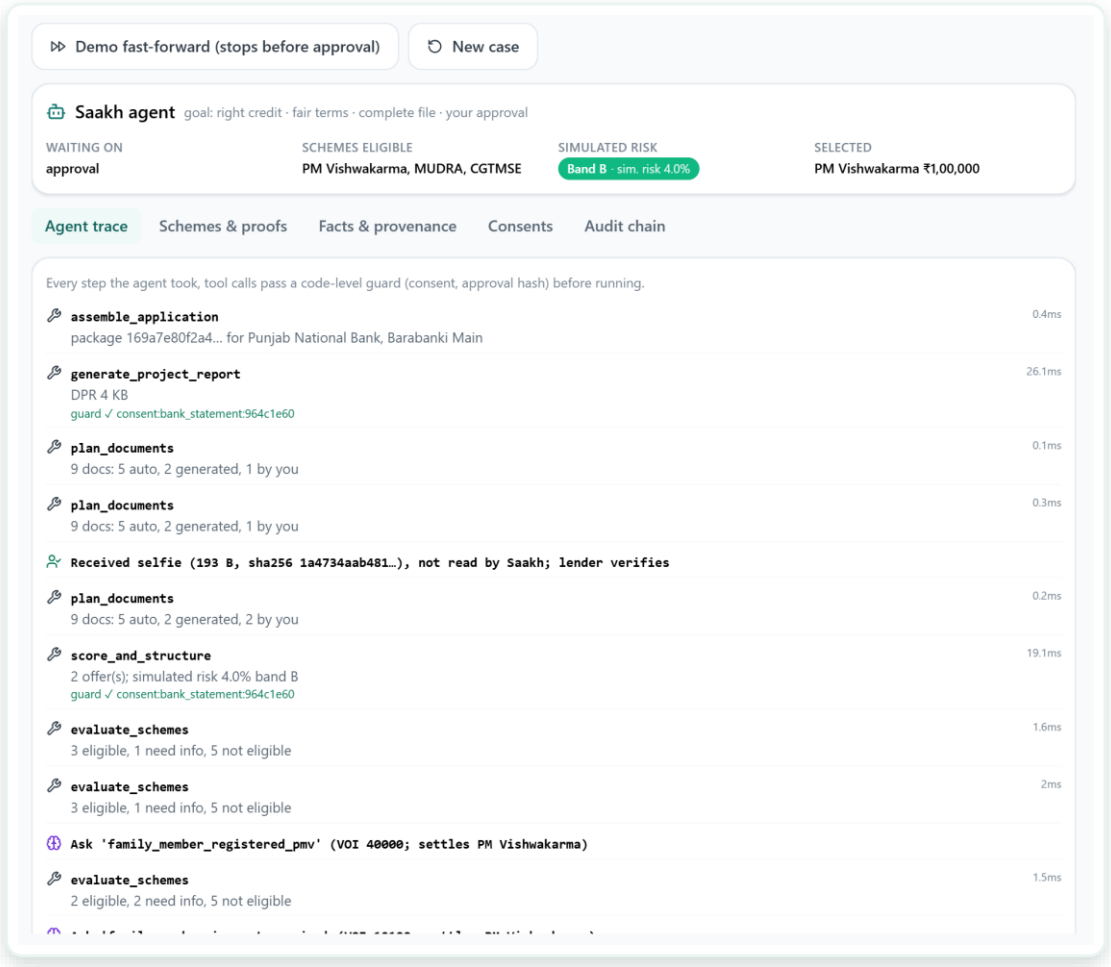
*Scales without rewrites: a new scheme is a YAML file with citations; the scorecard refits per lender; more Indian languages plug into the same parser design.*

## Jobs

**Direct:** café and business-correspondent operators paid per assisted journey; field verification for physical steps

**Indirect, the real lever:** every eligible-but-unreached unit that gets funded can hire, such as a second tailor, a fabricator's helper, or more milk sold to the cooperative

# Working today, end to end




**126**

automated tests, including concurrency and consent-lifecycle cases

**10**

scheme pathways with cited official sources and verification levels

**6**

synthetic personas, Hindi and English, voice in and read-aloud out

**6**

rounds of independent code review; every finding and verdict published

Also in the prototype: offline queue, JanAI Café assisted mode, branch console, governance view, SHG federation screening. Stack: FastAPI, React PWA, SQLite, numpy/scipy scorecard. Integrations without open APIs are simulated and labelled.

# What we will measure next



## Field research plan

- 15 to 20 owners across six business types, 3 to 5 branch managers, 2 to 3 loan agents, one SHG federation or café
- The organisers' six questions become hypotheses, each tied to a product decision it would change
- Branch time-and-rework log per file, to replace our cost assumptions with measured numbers
- Funnel and consent-decline rates measured by the product itself

### Next 30 days

Interviews and branch time study; refine questions and document weights

### By the finale

Pilot design with a branch and an SHG federation; measured rework numbers

### 2027

AA and ULI sandbox integration; scorecard refit on lender data with the same fairness gates; more languages

# Saakh turns trust already earned into formal credit



## Agentic

Plans, uses guarded tools, recovers from failure, and stops for the human at every consequential step



## Trusted

Consent, cited proof, exact-file approval and revocation, all enforced in code and tested



## Viable

Free for owners, paid by lenders who save on rework; scales on India's digital rails

**github.com/sumitmukherjee7/saakh**

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